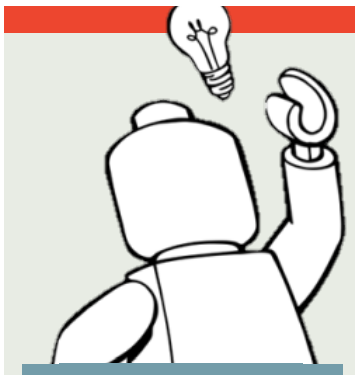


INFLUENCES

Globalisation

- Competitors outsourcing
- Competitors relocating plants to low cost countries
- Transportation
- Communications
- Global sourcing
- Inputs, facilities, distribution

Technology

- Leading edge injection moulding machines, serviced online from Switzerland
- Robotics (moving pieces, selecting pieces: robotic cranes, packaging)
- CAD/CAM
- E-commerce (suppliers, lego.com)
- RFID (Walmart, Target)
- 3d scanning tables (sculpted designs ie Police dog in Police Station)
- Mould prototyping and testing machines

Low tech

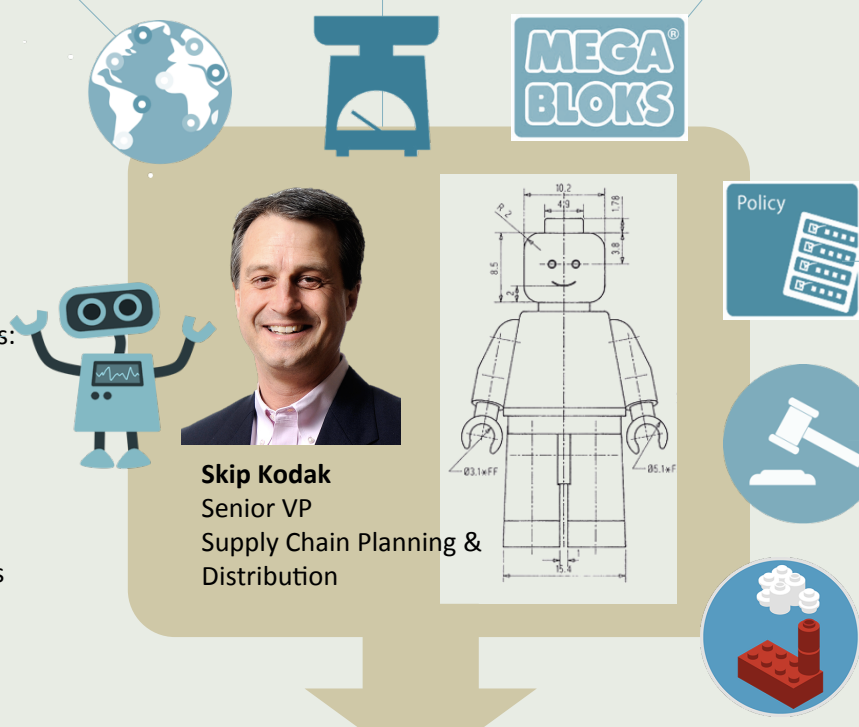
- Clay moulding for new pieces
- Scheduling and production KPIs (white board 7am Fri)
- Manual mould maintenance every 3 weeks including hand polish

Quality expectations

- Target zero recalls
- 0.00002% defects
- 0.0002 mm variance
- Interchangeable with all previous blocks
- The Apple of the toy business

Cost based competition

- Mattel, Hasbro relocated production to low cost countries before LEGO
- Megablocks competitor. Patent expired 2011
- 214 Mattel (Barbie) buy Mega Blocks for $460m. Sets are les than 50% price LEGO
- Significant discounts by Mattel and HASBRO and joint promotions with retailers



Skip Kodak
Senior VP
Supply Chain Planning & Distribution

Government policy
Legal regulation

EU Safety Directive

- Internal standards higher than Directive (even before legal)
- Tests of all the materials that are going into the bricks, the building instructions, the ink, whatever goes into the Lego box, to insure that we are fulfilling the directive.
- Process started a long time ago (when policy), to be ready to ensure that we have everything tested to [be able to] say that we are in line with the new directive.
- We don't bring any new material into the box unless they fulfill the new EU Toy Safety Directive's standard.

Environmental sustainability | CSR

Separate page

Designed in **1**

Billund, Denmark
by more than 180 designers of more than 24 nationalities

Moulds made in **Germany**
50k-300k Euro/mould

"It is our strategy to have production close to our core markets in order to secure short lead-time and world class service to our customers and consumers, and it has proven a successful strategy for us globally. "

Crude oil

- Saudi
- Kuwait

Processed

- Singapore
- Korea

Rubber

- Malaysia
- Thailand

0.5m cardboard boxes per yr

Made in **5** Factories

60 t/day
Store 420 t

Billund
Denmark

Kladno,
Czech Republic (2000)

Nyiregyháza
Hungary
(2015 new factory, tripling size)

Ciénega de Flores,
Mexico
(2008, add 48k sq m 2015)

Jiaxing
China
(start 2013, open 2017 165k m2)

2 Distribution centres

Jirny, Prague
Czech Republic
Managed by DHL
14 football fields
(competitors run 4)

Dallas,
USA
Managed by Deutsche Post

Shanghai,
China
Under construction

Sold in 130 countries

Offices

- 29 countries

Retail stories

- 5 countries

Online store

Moulding to distribution only **10 days**

Billund runs 24x7: every day, 60 tons of oil-derived granulate are pressed into Lego pieces by 768 moulding machines. 2m pieces produced per hour, packed into 424,000 boxes, held on 70km of shelving before being shipped out for decoration in Czech Republic, Mexico or for distribution from a DHL-run facility in the Czech Republic.

Production lines and block storage warehouse designed for maximum flexibility, and robotic automation.

All in all, Lego factories produce 33,000 bricks every minute, for a total of 16 billion bricks every year. The manufacturing facilities can make 3,000 different types of elements, including 300 million tires, and these pieces go into 37,000 Lego sets every hour.

Environmental sustainability | Corporate Social Responsibility

Sustainability

Smaller packaging

Lego goes after waste and wind

Hasbro certainly isn't the only toy maker making sustainable changes. Lego has worked for decades to eliminate PVC as well as phthalates from its toys, all of which no longer contain these substances. So while Hasbro gets props for working to remove PVC from its packaging, Lego actually has removed it from its products, something Hasbro has not committed to doing, although it is looking into alternative materials.

This year Lego is introducing a new series of smaller packaging boxes. The Danish company expects the change to reduce its annual consumption of cardboard by about 4,000 tons. Next year the cardboard used in the new boxes will carry FSC certification, meaning the Forest Stewardship Council guarantees it came from sustainable forests.

Wind farm

Over the next few years Lego's parent company, Kirkbi, is investing $547 million to build a wind farm off the coast of Germany. Lego and Kirkbi will control 32 percent of the Borkum Riffgrund wind farm, which amount to one of Germany's largest with 77 turbines when finished in 2015. By 2020, the company will contribute to the world at least the same amount of sustainable energy as the company consumes.

Lego says it has verified that every substance used in its products -- more than 2,000 raw materials -- comply with the European Union Toy Safety Directive, which imparts strict limits on what kinds of materials can be used in toys.

Waste recycling

Regarding waste, the company's approach is first to reduce the amount of waste it generates, and then reuse any waste material where possible, recycling almost everything that is left, said Morten Vestberg, sustainability communication manager for Lego.

"Today we recycle about 90 percent of our waste, and with zero waste as our long-term ambition we will continue to make progress on this agenda," he said.

It's also worth mentioning that Lego is one of the top 10 brands to land on the Reputation Institute's 2012 Global Corporate Social Responsibility Reputation list.

Ethical

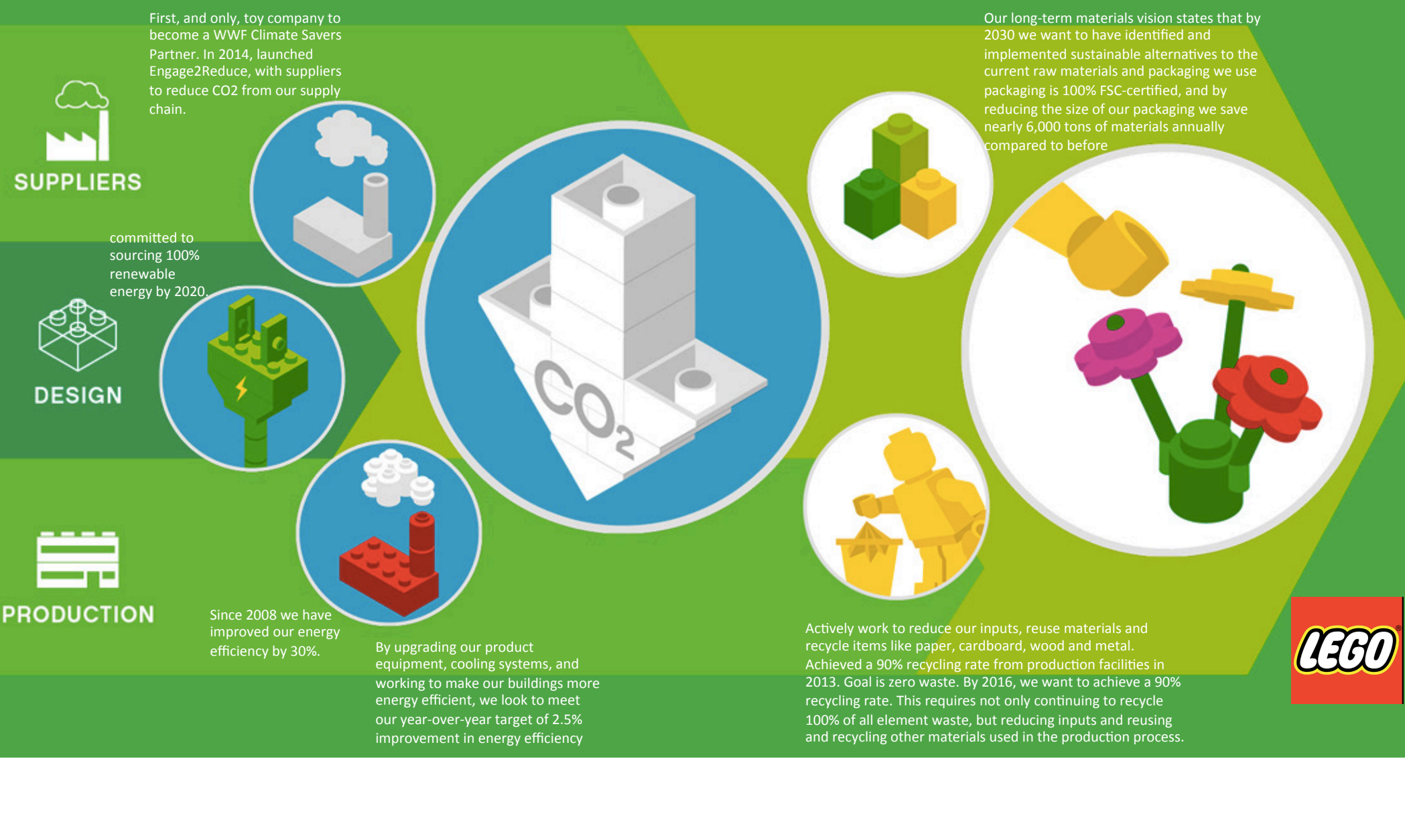
Lego's 57-year-old toy empire was built on plastic. But now the giant Danish toy company is investing millions into getting rid of it. By 2030, Lego bricks will no longer be made from ABS, the oil-based plastic in the 60 billion blocks the company makes each year.

"You could say that it's a logical place for us to find a way of reducing our environmental footprint," says Roar Trangbaek, press officer for Lego Group. "If you look at our CO2 footprint as a company, the majority of our impact comes from offscreen activities—basically what happens before we receive any raw materials in our factory.

The company is pouring 1 billion Danish krone, or around $150 million, into a new sustainable materials center that will open in 2016 and plans to add another 100 employees focused specifically on finding new materials for their toys and packaging. The classic bricks will likely still look and feel exactly the same when the company makes the shift; the material will still be plastic, just something that's not made from oil.

Quality

First and foremost, we have the same quality standards all over the globe. So it doesn't matter if a brick is moulded in Denmark or Hungary or in Mexico. It has to fulfill the same quality standards, so we ensure that the kids are getting the best possible physical experience with the bricks. And because we have the same standards around the globe, we can then train our quality people in different parts. Some are getting trained in the [Kladno], Czech Republic , in Nyíregyháza, [Hungary], others in Monterrey, [Mexico] and some are going to [Billund], Denmark. Normally what we do is, we have a lot of knowledge transfer going around the globe, so we share with each other the best practice to ensure that we are delivering the right quality. And also, our equipment is very standardized, so that also gives a lot of opportunities to do knowledge sharing around the globe for our quality people



TRANSFORMED RESOURCES

TRANSFORMING RESOURCES

	From (2004)	To: strategy (supply chain transformation
SKUs	• 5000 (1994) to 14,2000 (2004). • Product Development: introducing hundreds of new SKUs, more focused on product innovation than the impact on supply chain costs • 3,600 shapes required a mold, and a mold cost €50,000 on average, or up to €300,000 for complicated ones.	• Reduced to 5,000. Innovation through using a constrained number of SKUs that lower production costs. Increased per year, 9,000 (2011) • 130 new sets per year, constrained SKUs • By 2011, new products accounted for 50% of SKUs and 60-70% of sales each year. • New product development to less than 12 months • Required threshold "reuse rate" of each new SKU – that is, the portion of bricks that did not involve new shapes or colors
Colours	• 157 colours (2004) (up from 5 in 1980s)	• Reduced by half.
Suppliers	• 11,000 suppliers. • Frequently sourced unique (and therefore higher cost) materials, and did little to leverage its total buying power	• Significantly reduced plastic resin suppliers, signed longer term contracts reducing and stabilizing pricing, lowering costs and enabling better planning. • Promote sustainable supply chain, all suppliers and business partners must sign LEGO Group supplier code of conduct which covers human rights, labour rights, health & safety, anti-corruption and environment.
Customers	• Seen as children, serviced via small toy stores. • Little distinction in service policies between small mom and pops and the major retailers now accounting for majority of revenue	• Seen as children **AND** retailers. • "Let's rescue ourselves by making the retailers profitable ": emphasis on developing excellence in key account management. Annual customer satisfaction survey • dedicated people work with communities in different geographic regions, LEGO User Groups: regularly ask opinions, connect them to co-creation projects or specific development areas, and bring them together in conferences • Gains at retail, increasing typical shelf space from 9 linear feet in 2004 to 30 feet in 2009, retailer margins from 19% in 2004 to 30% in 2009, and inventory turns for retailers from 2x in 2004 to 5x in 2009.
Information	• produced what guessed would be demand, and at the same time, could not respond quickly enough.	
Supply chain	• In 2004, at least 10 years out of date: poor customer service, spotty availability of products. • Built for custom delivery to the smaller retailers, not changed as market changed to large retailers. • Other global competitors had changed supply chain, analyzing and optimizing every cost driver to provide just-in-time service to the new retail giants.	• Part of LEGO's sustainable competitive advantage • long-term strategy to have the packaging plants closer to their core markets in order to reduce delivery times.
Facilities	• Production processes and lines not set up to leverage its production scale, almost operating as a vast series of small, independent toy producers. • Long term planning a rarity, and fire drills common • Opened theme parks in UK, Germany and USA.	• Rationalized production lines, reduced number of products made on each machine, adding simplicity and reducing changeover costs. • Established more fixed production cycles, and better tied manufacturing to the rest of the supply chain. No longer acceptable to make manual changes in a molding machine without informing the finished-goods packing team. • new facilities built in Eastern Europe to better service the European market. 2004 consolidated most European warehouses and distribution centers (DCs) into one facility located in the Czech Republic saving 20% logistics costs (closing 11 DC and warehouses). Competitors have 3-4 Euro DCs. Savings reach 40% 2015 • Sold theme parks ($456m), 25% minor stake in management company • 30% expansion Kladno 2015. The high-bay warehouse has 72 km of shelf space with room for 180.000 storage boxes with LEGO elements. The automated warehouse cranes transport 600 boxes per hour. • 2013 Commence build China €100m, to open 2017 and service 70% Asia, building new DC Shanghai
Quality	• High brick quality and standardized stud spacing ensured all elements made after 1958 interlocked • molding tolerance of 0.002 millimeters	• Rebalanced quality focus: cost reducing ideas were not always seen as resisted as decreasing quality. • Documentation of existing processes provides a foundation for continuous improvement. • Project Polaris to document all business processes at LEGO (2.5 years). Global conventions, team roles, engagement processes to manage resistance to change. Communication and story telling key: A 3D model of the pyramid showing the process structure with different examples has been used together with big posters of the BPM vision and overview of the different steps of establishing process documentation. • Process documentation identified customer packing agreements that meant a lot of truckloads were only fifty-percent utilized. Lego was able to change some of the terms of its sales agreements to eliminate inefficient handling and distribution practices. • In addition to the Business Process Management function, the LEGO Group has a LEGO Continuous Improvement function which aims to create a culture of improvement of the LEGO Group. • all conventions for process documentation are adapted to meet quality requirements (ISO 9001).
Human resources		• Fire 7 senior manufacturing executives • Challenge to find skilled workers in Czech republic, LEGO needed to conduct extensive training • Doubled revenue per employee 04-14
Technology		• LEGO website, brings together groups through games, stories, activities, and experiences. By 2009, topped the list of family and children's internet sites. LEGO Clubs for children had a membership of 3.2 million and a dedicated website • See logistics re web based logistics system with DHL • Highly automated robotic molding, decorating, picking and packing • Made mindstorms software open source to allow hacking

		From (2004)	To: strategy (supply chain transformation
TRANSFORMING RESOURCES	Outsourcing	• Outsourced to Flextronics (manuf). Reduced headcount to 3,000 from 8,300. Products characterized by multiple labor-intensive decorating and assembly or packing processes will be outsourced because their high degree of uniformity and large production volumes render outsourcing profitable, partly because the packing of these products is very wage intensive. • Close US based factory in Enfield, Connecticut and relocate to Flextronics' in Mexico. (300 jobs) • Distribution relocate to Mexico • Billund retains technically more demanding products (LEGO Technic and BIONICLE). • LEGO System products will relocate from Billund to Flextronics' facilities in Eastern Europe. (90/1,200 jobs gone) • Flextronics takes over operations at the LEGO Group's factory in Kladno, the Czech Republic. • Outsource management of distribution centre in USA and Czech Republic • Outsource European distribution	• inventory accuracy fell, lifetimes of molds shortened due to lower maintenance levels, purchasing costs rose instead of decreasing. 12 week lead time • The coordination and control at a global scale became very complex leading to conflicts • Identified their main issues: lack of proper documentation and standardization of business processes, the need of proper communication lines and efficient organization structures, to be able to manage a global and multicultural company. • 2009 cancelled Flextronics contracts, moved most production back in-house, to its sites in Eastern Europe, Mexico, and Denmark. "it was a big mistake to consider manufacturing as non-core." • invested hundreds of millions of dollars to rebuild its own manufacturing capabilities, still used Chinese contractors for electronics and decoration. • Cutting-edge molding technology and high automation: 3 people handle 64 machines molding machines (China,3 people per machine) • decoupled molding from packaging and ordering, reduced the inventory of finished goods, and increased inventory of common elements. • Now outsource only where lack in-house capabilities like the electronics for LEGO MINDSTORMS or some complicated manual decorations • Distribution outsourcing very successful
	Flexibility		• Can mould, decorate and package within 10 days to respond to customer demand changes which are evident in global DCs. Competitive advantage in market.
	Cost	• No cost control disciplines • management did not see impact of SKU explosion (variety) on design, manufacturing, servicing of retailers, forecasting, and managing inventory.	• Developed a cost matrix that calculated for designers the high cost impact on the supply chain of changing product colors and shapes.
DISTRIBUTION	Inventory management	• fill rate to customers [proportion of demand delivered without delay from stock on hand], untracked, anywhere between 5% and 70% • major customers frustrated by stock-outs and slow-moving inventory	• JIT for the retailer. Very seasonal sales needs great flexibility in manufacturing processes and large scale distribution centre to store product • can take more risks on inventory molding, and now nearly level-loaded all year round even though seasonality remains massive (with about 50% to 60% of our sales happening during the Christmas weeks). Mold common pieces throughout the year and then chase 30%, depending on demand, and pack as late as possible
	Logistics and distribution centres	• 26 logistics providers	• Cut to 4. • New DC in Eastern Europe to take advantage of lower costs than in Germany and other major markets – before competitors used Eastern European logistics. • Working much more closely with retailers on joint forecasting and logistics planning • Needed to change shipment scheduling to improve load consolidation with outsource DC and logistics. Lego and DHL develop web-based transportation management systemto tender loads to carriers, optimize loads, and route shipments, taking into account such factors as national holidays to ensure that Lego meets its customers' delivery requirements. Lego and DHL decided to build their own solution to accomplish all this after careful review of the existing packaged solutions.
BUSINESS PERFORMANCE	Revenue	• Sales dropped 30 percent in 2003 and 10 percent more in 2004, to $1.35 billion world wide • The Lego City product, which once accounted for 13 percent of the company's total revenue, plummeted to just 3 percent	• "LEGO products are now sold by pretty much anybody who claims to be selling toys, and we are envied for our retailer relationships." • 2010, sales jumped 37% to DKK 16 billion • Revenues grew faster than the toy market and continued to dominate the construction segment with 80% market share. • Revenue triples between 2007 & 2013
	Gross margin		• 2007 gross profit margin 71%
	Profit	Losses 4/7 years from 1998 through 2004.	• Return to profitability 2005. Save US$100 million • Inventory turns improved by 12%. • From being well behind the competition in supply chain excellence to what it believes is a slight competitive advantage. • 2010 net profit soaring 69% to DKK 3.7 billion. • Net profit margin up from 4% (in the early 2000s) to 30%, far above the industry average

"At least 50% of the LEGO Group's success is due to the optimization of our complex operating model – the system of factories, distribution centers, inventory, packaging, and shipping. We mold about 30,000 bricks every minute of the day, 24 hours a day, in 6,000 to 7,000 variations. They need to be sorted to go into 350 retail SKUs that have to be packaged in their boxes and delivered with precision. If you don't meet the 7am cross-docking window at Wal-Mart, they tell you to come back the next day. " **CEO Knudstorp**

The LEGO Brand Framework

Mission	Inspire and develop the builders of tomorrow	
Aspiration	Globalize and innovate the LEGO system-in-play	
Promises	**Play Promise** Joy of building. Pride of creation	**Partner Promise** Mutual value creation
	Planet Promise Positive impact	**People Promise** Succeed together
Spirit	Only the best is good enough	
Values	Imagination - Creativity - Fun - Learning - Caring - Quality	